

Augmented Kitchens: XR-Enabled Communication in Active Domestic Workspaces

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Abstract. Human communication in active domestic spaces, such as kitchens, is inherently embodied, spatial, and multisensory. Yet current communication tools remain static, designed for seated and screen-bound interaction. This paper investigates how extended reality (XR) and ubiquitous computing can augment perception, presence, and communication in such dynamic contexts. Through a three-stage mixed-methods study, comprising observations of fifteen online cooking workshops, seven co-design ideation sessions with 42 participants, and a VR prototype evaluation with professional cooks, we identify key spatial, sensory, and social challenges of communicating in motion and translate them into three XR artifacts: a real-time kitchen-to-kitchen portal, a 3D asynchronous messaging system, and an immersive ambient cooking environment. A Continuous Fuzzy Kano analysis reveals that privacy controls emerge as near-essential requirements. By reframing the kitchen as a testbed for embodied, XR-enabled communication, this work extends human augmentation toward social, perceptual, and communicative dimensions.

Keywords: Extended Reality (XR); Virtual Reality; Spatial Computing; Embodied Communication; Human Augmentation; Domestic Environments; Co-presence; User-Centered Design

1 Introduction

Human communication extends beyond words; it is embodied, spatial, and richly multisensory. Yet most communication systems remain designed for stationary, screen-bound contexts, poorly suited for the dynamics of movement, material interaction, and sensory engagement. Emerging paradigms in *ubiquitous computing* and *extended reality (XR)* present new opportunities to augment human perception, expression, and social awareness in these complex settings. The ambition to amplify human capability through technology traces back to Engelbart’s vision of *Augmenting Human Intellect* [4]. In the decades since, research in human augmentation has evolved to encompass physical, cognitive, and perceptual enhancement through digital means. This paper situates itself within that lineage, focusing

on how ubiquitous and XR-based systems can extend human communication capabilities, supporting awareness, empathy, and co-presence across spatial and sensory boundaries.

We examine this question in what we term *Active Workspaces* [11]: environments such as kitchens, workshops, or studios where people engage in overlapping physical and social tasks. These spaces challenge existing communication paradigms because they require users to interact while multitasking, moving, or manipulating tangible objects, yet conventional touch-based or screen-focused systems often disrupt rather than enable these embodied exchanges. The kitchen, in particular, serves as a compelling context; it is globally familiar, culturally meaningful, and inherently multisensory, representing a microcosm of human coordination where communication unfolds through gesture, proximity, sound, and shared attention. Remote communication in such contexts is increasingly common through virtual workshops, live-streamed tutorials, and distributed family activities, making the kitchen an ideal environment for examining how embodied, hands-free, and multisensory communication can be technologically augmented.

Prior work has explored how domestic environments can be augmented through mixed and augmented reality. Knierim et al. [12] investigated how users envision AR augmentations in everyday home life, identifying themes of assistance, enhanced perception, and social activity, primarily focused on human–environment interaction. In contrast, our study turns toward human–human communication within these same everyday environments, reframing augmentation as a social, perceptual, and embodied phenomenon. By situating kitchens as live, sensorily intense workspaces, we extend domestic AR research from object- and task-based augmentation to communicative augmentation and multisensory co-presence.

This study explores how spatial computing and XR technologies can *augment communication* in active domestic spaces. Unlike prior research on smart kitchens, which has largely focused on automation, monitoring, and assistive functionality, we approach technology as a means of extending human presence, perception, and social connection. Through a three-stage mixed-methods study, we investigate real-world communication breakdowns, user-driven design visions, and XR prototype evaluations in kitchen environments. The first stage observes 15 online cooking workshop sessions, Figure 1), which served as natural experiments in remote co-presence, revealing breakdowns and adaptations in embodied communication. The second stage comprises seven ideation workshops where 42 participants envisioned how ubiquitous and XR technologies could enhance awareness, coordination, and expression in the kitchen. The third stage evaluates three VR prototypes with professional cooks using a Continuous Fuzzy Kano analysis to assess user requirements. Our research addresses three central questions:

1. What challenges do users face when communicating in active, embodied domestic environments?
2. How can ubiquitous computing and XR augment perception, awareness, and presence in such contexts?

3. What design opportunities emerge for multisensory, context-aware, and socially meaningful augmentation of communication?

We present four observation-based challenges and four ideation-based design themes (Figure 2) that collectively articulate a framework for *communication augmentation*, grounded in empirical observation and validated through XR prototyping. By treating the kitchen as a testbed for ubiquitous and XR-enabled communication, we contribute to the broader discourse on human augmentation, demonstrating how technology can enrich human experience, empathy, and connection in the fabric of everyday life through emerging ubiquitous and spatial computing.



Fig. 1: Key challenges observed during the cooking workshops. Visibility issues appear in multiple sessions due to limited camera angles and phone orientation. In some sessions, tutors used extra cameras or assistants to demonstrate techniques, while some participants cooked outside the kitchen or chose only to observe.

2 Related Work

This paper situates itself at the intersection of three overlapping domains: Smart Homes and Internet-of-Things (IoT), Extended Reality (XR) and spatial computing for embodied interaction, and design research on communication in domestic environments. Within these domains, we focus specifically on how ubiquitous, spatial, XR technologies can *augment human communication and perception* in active workspaces—environments where users are physically and cognitively engaged with tasks, tools, and materials. While prior studies have explored automation and task support within smart kitchens, comparatively little research has examined how such environments might serve as testbeds for augmenting

human-to-human communication through spatial and multisensory augmentation. In this paper, we use the term spatial computing to denote systems that sense, interpret, and respond to human activity in physical space—through sensors, projection, or XR interfaces—aligning with contemporary definitions of ubiquitous and embodied computing in HCI.

Researchers have approached the Smart Kitchen from diverse perspectives. Eom et al.[5] introduced a system using smart Tupperware as an IoT solution to automate shopping lists and provide recipe suggestions, while Balaji et al.[2] proposed integrating image recognition into kitchen wardrobes for similar purposes. Oliver et al.[18] developed the *Ambient Kitchen*, a lab-based replica incorporating RFID sensors, projectors, and cameras to support ubiquitous computing research in cooking contexts. Patterson et al.[23] and Pham et al.[24] leveraged RFID for motion recognition, while Hilliges et al.[10] explored mid-air interaction through depth-sensor cameras. Similarly, Panger[21] and Hashimoto et al.[9] used vision-based systems for recognizing gestures and cooking actions. These works illustrate the growing ambition of ubiquitous computing to embed digital intelligence into domestic activity, though most focus on efficiency and automation rather than on augmenting perception, communication, or co-presence.

Ju [11], in her doctoral thesis, synthesized work from concept kitchens such as *CounterActive* and *Origami Desk* to propose “active workspaces” as sites where tangible and digital interaction converge. Building on this foundation, Bonanni’s *CounterIntelligence* project [14] employed projection-based guidance systems to assist cooking processes, while Kosch et al.[13] designed a Kinect- and projector-based kitchen for cognitively impaired users, showing how spatial augmentation can enhance coordination and independence. Recent work in the Augmented Humans community has expanded this perspective through the concept of *Assistive Augmentation*—interfaces that integrate seamlessly with body and behavior to extend perception and action [17]. Nanayakkara et al. proposed a two-dimensional design space (bodily integration $\times$ ability), framing augmentation as a continuum from assistive to superhuman. This framing helps position kitchen communication systems as *assistive augmentations*—supporting not only efficiency but also sensory and social expansion within everyday routines.

Knierim et al.[12] examined AR interfaces in home environments, extending domestic interaction into augmented visual layers. The *Telelife* project [19] speculated on future domestic XR applications, envisioning connected kitchens with enhanced remote communication capabilities. Similarly, the *CoDine* studies [32,31] proposed multisensory co-dining systems that support remote family meals, illustrating how XR can augment both sensory perception and social connectedness. Beyond cooking assistance, Mennicken et al.[16] embedded interactive displays into kitchen counters for *First Person Cooking*, and Terrenghi et al.[27] explored social recipe sharing via the *Living Cookbook*. Patibanda et al.[22] further introduced the notion of *intra- and inter-corporeality* in augmented human systems, emphasizing shared bodily control and mutual coordination across multiple users. This perspective aligns with our focus on co-present kitchen interactions, where communication involves not only cognitive and sensory exchange but also

bodily synchrony and spatial negotiation. Together, these projects demonstrate how XR and spatial technologies can be leveraged to augment not only task performance but also embodied social experience.

Further research has investigated learning, assistive, and conversational augmentation in kitchen contexts. Preston et al.[25] transformed the kitchen into a smart learning environment for language acquisition, combining sensor-based feedback with everyday activity. Conversational agents such as *Foodie Fooderson* [1] and *SmolKat* [7] aimed to assist with nutrition, meal preparation, and task sequencing through speech-based guidance, while Hamada et al.[8] developed *Cooking Navi* to manage multi-recipe coordination. These studies highlight the growing role of multimodal, context-aware systems in augmenting cognitive and behavioral capabilities during cooking. However, most remain focused on instrumental or assistive augmentation, leaving the social and communicative dimensions underexplored. The perception and acceptance of augmentation itself have also become a growing concern in HCI. Villa et al.[28] conducted a mixed-method study on how people perceive augmented humans across sensory, motor, and cognitive dimensions, showing that social attitudes toward augmentation depend strongly on perceived fairness, safety, and authenticity. Their findings underline that augmentation is not merely technical but inherently social—a view our work extends by exploring *communicative and relational augmentation* in shared domestic spaces.

Designing technologies for domestic environments also requires sensitivity to their spatial and cultural dynamics. Kitchens have long been central to design research, from Lillian Moller Gilbreth’s early “kitchen work triangle” optimizing efficiency through spatial layout [26], to contemporary inquiries emphasizing collaboration, emotion, and well-being. For example, Paay et al.[20], in *Cooking Together*, used proxemics and F-formation theory to analyze co-located and remote coordination in kitchen activities. Qi et al.[26] synthesized quantitative and qualitative findings to evaluate how user-centered kitchen design contributes to well-being under the PERMA model. Wang et al.[30] applied space syntax analysis to study aging-in-place, identifying how layout influences accessibility and daily performance. Mathis et al.[15] recently examined how AR systems intersect with *everyday life challenges*, arguing for contextual, situated, and socially grounded augmentation rather than purely functional enhancement. Their insights resonate with the domestic, improvisational settings of our study, where augmentation operates within the rhythms and constraints of everyday practices. These works broaden the understanding of domestic design from efficiency toward inclusivity, sociality, and emotional resonance—key factors for human-centered augmentation.

Despite this substantial body of work, few studies examine synchronous, embodied communication where users must coordinate, speak, and move simultaneously in sensory-rich domestic contexts. Our three-stage mixed-methods inquiry addresses this gap by (1) characterizing communication challenges in motion, (2) identifying user needs and design opportunities for XR-enabled augmentation,

and (3) evaluating XR prototypes that extend human perceptual, social, and communicative capacities beyond efficiency and automation.

3 Methodology

This study employs a three-stage mixed-methods methodology to investigate how ubiquitous computing and extended reality (XR) technologies can augment communication practices and embodied interaction within kitchen environments as active workspaces. As illustrated in Figure 2, the research was structured around three interconnected stages: (1) observational fieldwork conducted during 15 online cooking workshops, (2) a series of 7 ideation workshops designed to elicit user needs and speculative design opportunities, and (3) a VR prototype evaluation with professional cooks using a Continuous Fuzzy Kano analysis to assess user requirements. The first stage identifies the essential communication challenges users face in dynamic domestic contexts. These findings informed the second stage, which explores how emerging modalities such as ubiquitous, spatial computing and XR might augment embodied communication. The third stage translates these insights into three XR prototypes, evaluating how they meet user needs and expectations. Through this iterative process, we ground speculative design in empirical observation, generating a framework for embodied, multisensory, and context-aware communication in domestic environments.

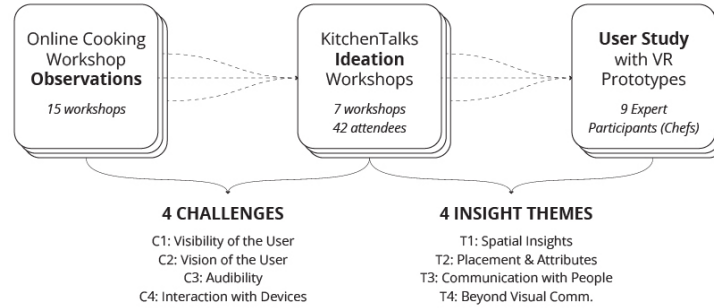


Fig. 2: Overview of the three-stage KitchenTalks study: 15 cooking workshop observations yielding four challenges (C1–C4), seven ideation workshops with 42 participants yielding four design themes (T1–T4), and a VR prototype evaluation with 9 expert participants.

3.1 Stage 1: Cooking Workshop Observations

To explore communication challenges in the kitchen, we observed a series of online cooking workshops organized by a non-profit initiative gathering people to cook, share recipes, and discuss cultural themes. One researcher joined each session as

a silent observer with prior approval from the organizers, documenting verbal and non-verbal exchanges through detailed field notes. Each session involved three roles: *the tutor*, who conducted cooking demonstrations; *the organizer*, who managed audio-visual coordination and participant moderation; and *the attendees*, who joined from their own kitchens (5–12 per session), sometimes cooking along, sometimes observing. After each workshop, a 30–45 minute debrief consolidated findings and identified recurring issues. Using an inductive thematic coding approach, we categorized communication challenges such as device placement, visibility, and multitasking limitations, which directly informed the ideation phase and provided empirical grounding for subsequent XR augmentation opportunities.

3.2 Stage 2: Ideation Workshops

Seven online ideation workshops were conducted with 42 participants, recruited through an open call including everyday kitchen users and four professional cooks. Workshops were held via Zoom, using Miro for collaborative tasks, and structured around three phases. *Warm-up*: Participants sketched their kitchen layouts using provided templates, identifying areas for activity, communication, and ideal technology placement. This activated spatial memory and revealed physical constraints affecting communication flow. *Reflection on daily habits*: Participants described their cooking practices, task rhythms, and communication behaviors, such as whom they communicate with, which devices they use, and under what conditions, deepening understanding of multitasking demands and informing design considerations for context-aware and hands-free augmentation. *Group ideation*: Participants conceptualized speculative tools for kitchen communication, exploring how spatial computing and XR could enhance perception, co-presence, and awareness, revealing both functional needs and imaginative visions for future communication augmentation.

3.3 Stage 3: User Study through Virtual Prototypes

In the final stage, we designed and built three VR prototypes based on findings from the previous stages. Artifacts were modeled in *Blender* and integrated as interactable objects into a virtual kitchen scene in *Unity*. The *Record3D* iOS app was used to capture high-resolution 3D video messages via the device depth camera, and a Unity package *Portals for VR* [6] was used to implement portal interactions.

To evaluate the design insights emerging from the KitchenTalks ideation workshops, each prototype addressed a distinct communication dimension identified across the four themes: immersive ambient experience, real-time remote presence, and asynchronous spatial messaging. Virtual reality was chosen as the prototyping medium because it allowed us to simulate situated, kitchen-embedded interactions that would be technically complex and costly to deploy in real environments at this stage, enabling participants to experience and evaluate speculative communication artifacts in a realistic spatial context. In the user study, participants were first briefed on the study purpose, the experience flow, and potential risks. They

were then introduced to the basic interactions in VR and asked to freely explore the virtual kitchen for five minutes to familiarize themselves with the environment. After this familiarization period, participants experienced each prototype individually in the following order: (1) Immersive Kitchen, (2) Kitchen Portal, and (3) Kitchen Annotation. Following the experience, participants completed a Continuous Fuzzy Kano Model questionnaire to evaluate the prototypes and their features [33,34]. We employed this approach and analysis method by considering the characteristics and conditions of the prototypes, guided by the assessment framework proposed by Violante and Vezzetti [29].

(1) *Immersive Kitchen*: Derived from ideation workshop suggestions (e.g., tools that boost mood, play context-aware music, or enhance cooking with augmented sounds), this prototype places the user in a pizza-making scenario where the environment responds dynamically: themed music plays, ambient spotlights activate, and a projected backdrop of an Italian coastal town appears behind the counter, creating a multi-sensory immersive cooking atmosphere.

(2) *Kitchen Portal*: Inspired by the portal metaphor, this prototype enables real-time kitchen-to-kitchen communication through a fixed in-situ device. It supports parallax view for enhanced depth and presence, gesture and voice control for hands-free operation, and transmission of spatial cues such as pointing and highlighting between remote spaces. In the multi-user VR experience, participants appeared as Meta Avatars, could see each other’s gestures and the remote kitchen through the portal, and were able to reposition and reshape the portal within their kitchen space.

(3) *Kitchen Annotation*: Inspired by fridge magnets, this prototype supports asynchronous communication through 3D-recorded messages that appear pinned to the fridge door. Users can view messages in 3D at their convenience and manage them by hiding, moving, or saving them. For the study, six short 3D video messages were pre-recorded, covering everyday scenarios such as requesting a meeting, asking for a recipe, and noting a delivery.

3.4 Analysis

Two complementary analyses addressed the spatial and narrative dimensions of the data. Kitchen layout drawings were examined using a custom clustering method to identify common spatial relationships, while textual data from Miro boards and discussions underwent reflexive thematic analysis [3], capturing both physical configurations and conceptual patterns relevant to communication augmentation.

Kitchen Plan Analysis Kitchen plans varied widely in size, shape, and structure. We developed a clustering approach emphasizing relational proximity among activity zones, devices, and communication points (Figure 3), allowing comparative interpretation of how spatial arrangements influence communicative

behavior. The synthesized relationships were then visualized as 4D diagrams (Figure 4), representing how work areas, device positioning, and communication flows intersect across time and activity.

Thematic Analysis Following Braun and Clarke’s reflexive thematic analysis framework [3], two authors independently coded the qualitative data through six iterative phases, comparing and refining codes in collaboration with HCI researchers. Key themes included embodied communication barriers, adaptive practices, and speculative augmentation concepts, forming the basis for a design framework of *ubiquitous and XR-enabled communication augmentation* in domestic environments.

User Study Survey Analysis For the survey, we adopted a hybrid quantitative approach grounded in the Continuous Fuzzy Kano Model (CFKM) [33]. Category assignment was inspired by the Analytical Kano model [34] and operationalized through Euclidean distance-based classification. Specifically, we computed the average functional and dysfunctional values for each feature and measured their respective distances to the four ideal Kano categories: Attractive (A), Performance/One-Dimensional (O), Must-Be (M), and Indifferent (I). The questionnaire included assessment items for each overall prototype (Table 2), as well as their specific features (Table 3 & Table 4). Ultimately, this analytical framework classifies user requirements based on the extent to which each feature impacts overall user satisfaction.

4 Findings

This section presents communication challenges and design opportunities emerging across the first two stages of our study. The cooking workshop observations revealed everyday challenges in active domestic environments, including misaligned attention, occluded visibility, and fragmented coordination, while the ideation workshops built upon these insights to explore how XR technologies could address them through speculative design.

The synthesis of both stages yielded four overarching thematic areas: **(1) Spatial Insights**, showing how kitchen layouts and movement patterns shape perception and interaction; **(2) Placement & Attributes**, analyzing where and how devices are positioned and perceived; **(3) Communication with People**, examining the social and embodied dimensions of coordination; and **(4) Beyond Verbal Communication**, exploring speculative, multisensory, and ambient modes of interaction. Together, these themes frame the kitchen as a testbed for *communication augmentation*: technologies that extend perception, awareness, and presence beyond efficiency, toward expressive, socially attuned interaction (see Table 1).

4.1 Cooking Workshop Observations and Challenges

Across workshops, four recurring challenges—*visibility*, *vision*, *audibility*, and *interaction with devices*—revealed how domestic cooking spaces constrain commu-

Table 1: Summary of findings from cooking and ideation workshops

Study Phase	Key Challenges and Themes
Cooking Workshop Observations and Challenges	Challenge 1: Visibility of the user — Limited camera angles and device placement hindered users’ ability to be seen clearly. Challenge 2: Vision of the user — Small screens and distant placement of devices made it hard to follow others. Challenge 3: Audibility — Background noise from kitchen activities disrupted verbal communication. Challenge 4: Interaction with devices — Wet or busy hands and unfamiliar interfaces caused usability issues.
Ideation Workshop Findings	Theme 1: Spatial Insights — Diverse kitchen layouts showed mobile device use and spatial separation of work and communication zones. Theme 2: Placement & Attributes — Participants requested mobile, wall-mounted, or durable communication tools for kitchen environments. Theme 3: Communication with People — Tools should support real-time interaction, ease of expression, and group coordination. Theme 4: Beyond Verbal Communication — Envisioned expanding kitchen communication through multisensory and ambient experiences. Theme 4.1: Communication with Taste — Speculative use of edible 3D-printed messages and cultural food rituals for emotional connection. Theme 4.2: Augmented Cooking Experience — Use of ambient sound, video, or light to enhance immersion and communication. Theme 4.3: Immersive and Ambient Experience — Smart kitchens could amplify cooking sounds for engagement and awareness.

nication. These breakdowns are not merely usability deficits but invitations for *augmentation*: ways to extend perceptual, auditory, and bodily capacities under realistic working conditions. Camera framing that dynamically tracks bodies and tasks (*perceptual augmentation*), audio systems that separate appliance noise from speech (*auditory augmentation*), and input modalities that remain usable with wet or occupied hands (*multimodal augmentation*) all point to communication as a co-adaptive process between humans and tools. Participants’ improvised solutions—using dual devices, mounting phones on jars or tripods, asking others to act as mediators—already hint at the need for *distributed* augmentation: multiple cooperating sensors and displays that sustain mutual awareness as people move through space.

Challenge 1: Visibility of the user Participants frequently vanished from view when turning toward the sink, leaning over a pot, or stepping aside to retrieve ingredients. These moments highlighted the fragility of single, static viewpoints

in active spaces. Instructors often asked, “Can we see what you’re doing?” while participants awkwardly adjusted devices mid-task. Such breakdowns indicate a need for *multi-perspective capture*—wearable cameras, steerable tripods, or depth-aware framing that automatically centers relevant gestures or artefacts. Beyond solving occlusion, visibility becomes a lens for *embodied legibility*: designing systems that maintain the readability of human action as a key communicative channel.

Challenge 2: Vision of the user While visibility concerns what others see, vision concerns what participants themselves can perceive. Many positioned devices away from the main work area for safety or space, leaving remote collaborators literally out of sight. Participants would lift dough-covered hands toward screens to check cues, then resume work with delayed feedback. This points to a demand for *proximal, glanceable, and projective* visual channels that keep partners “in the corner of the eye.” Under an augmentation lens, this includes *ambient or situated displays*—for instance, low-resolution projections on cabinets or head-mounted overlays—that preserve social presence without fragmenting attention between screen and workspace.

Challenge 3: Audibility Sound emerged as a constant negotiation. The roar of blenders, boiling pots, and echoing kitchens routinely masked speech, while remote participants struggled to locate who was speaking. Some muted appliances or raised voices; others relied on improvised hand signals. These coping strategies underscore the need for *context-aware audio augmentation*: directional microphones that track speakers, adaptive noise suppression that distinguishes between utensil clatter and voice, or lightweight push-to-talk gestures triggered by motion or gaze. Augmentation here is less about perfect fidelity than maintaining conversational continuity amidst acoustic turbulence.

Challenge 4: Interaction with devices Finally, interaction breakdowns exposed the embodied friction of managing tools mid-task. Wet or flour-covered hands made touchscreens unreliable, while interface icons were too small to manipulate at a distance. Participants resorted to elbows, knuckles, or requests for help from family members—revealing how bodily constraints shape communication. Under an augmentation frame, this invites *hands-free interaction grammars*: voice or foot input for simple controls, proxemic gestures near work zones, and micro-automations such as auto-muting when appliances start or spotlighting the cook during key steps. Such features would transform device handling from an interruption into an integrated rhythm of motion and attention.

Taken together, these four challenges delineate the lived complexity of “communicating in motion.” They make visible the latent design space where ubiquitous sensing and XR technologies could enhance not just task efficiency but the perceptual, auditory, and expressive synchrony that underpins domestic collaboration.

4.2 Ideation Workshop Findings

Building on the empirical insights from the cooking workshop observations, the second phase of the study invited participants to envision how emerging technologies could address those communication challenges. Through seven ideation workshops, we explored user-driven proposals that reframed kitchen interaction as a space for perceptual, spatial, and social augmentation—transforming everyday coordination into opportunities for expressive and embodied communication.

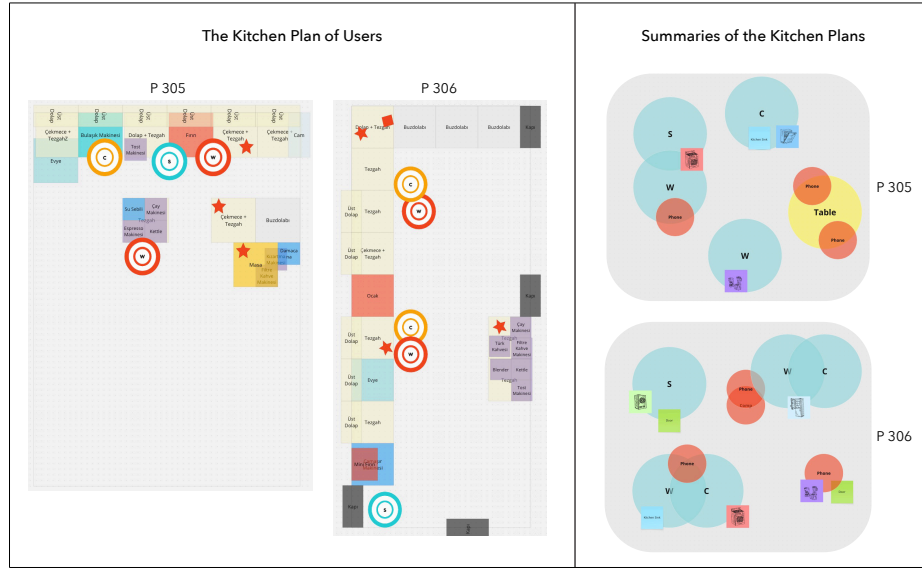


Fig. 3: Two participants’ kitchen plans (P305 and P306) and their abstracted summaries. Original layouts (left) show working zones and device placements, while the summaries (right) cluster key functional areas, Work (W), Storage (S), and Communication (C), to visualize spatial relationships across diverse kitchen configurations.

Spatial Insights Participants’ spatial diagrams and discussions revealed that the kitchen’s geography fundamentally shapes communication—where it happens, how it flows, and where it breaks. Conversations naturally gravitated toward *tables and shared surfaces*, which served as informal command centers for social interaction, while the most intensive physical work clustered around counters, stoves, and sinks. This polarity between social and functional zones points toward distinct augmentation opportunities: *situated, low-footprint display zones* near tables to facilitate shared awareness, and *lightweight, peripheral overlays* near work anchors to sustain coordination without disrupting motion. The

recurring insistence on keeping workspaces “clean” or “uncluttered” underscores the preference for *ambient*, non-dominant XR visuals or sounds that hover at the perceptual edge, surfacing only when context demands it.

Diagram 1: Table-centric clusters emphasize that communication thrives around social anchors. Participants imagined “co-present social augmentation”—shared glanceable displays or subtle lighting cues indicating who is speaking or seeking attention—to preserve conviviality and turn-taking without the cognitive burden of device management.

Diagram 2: Overlaps between current phone placements and proposed solution areas show how augmentation can *piggyback on existing ecologies*. Rather than introducing alien devices, participants preferred enhancements; mounts, docking spots, or wearables; that extend their habitual rhythms. This reflects a user-driven logic of continuity: augmentation that grows from lived practice rather than overwriting it.

Diagram 3: Avoidance of placing devices directly in high-intensity work areas reflects a concern for safety and fluidity. Participants proposed technologies that *reach into* those zones—steerable cameras, projected pointers, head-worn micro-overlays—so that guidance and feedback remain accessible while hands stay free. This frames augmentation as a distributed presence that orbits rather than occupies the workspace.

Diagram 4: The omnipresence of phones in all diagrams positions them as *mobile augmentation hubs*. Participants envisioned the phone not as a fixed screen but as a “roaming coordinator” orchestrating other peripherals—clip-on microphones, auto-tracking cameras, ambient projectors—creating adaptive *situational assemblages* that reconfigure as the activity unfolds.

Placement & Attributes Participants translated their daily irritations—tiny screens, slippery devices, muffled speakers—into concrete design demands: *hands-free control*, *glanceable feedback*, *directional sound*, and *ergonomic camera mobility*. These are not mere wishlist items but expressions of lived bodily negotiation with tools. The “laptop + phone” workaround observed among tutors, where one device handled visuals and another managed voice, prefigures a *distributed augmentation paradigm*: a multi-node system that mirrors human division of attention. During ideation, participants refined this concept through proposals for wall-mounted displays, context-sensitive tracking views, and flexible camera mounts. Collectively, these envision the kitchen as an orchestrated communication ecosystem rather than a single smart appliance—where each node contributes to seamless continuity as users circulate through different zones.

Solution Approach: Always with Me / Using Additional Cameras Participants repeatedly voiced the desire for technology that “follows me” and maintains

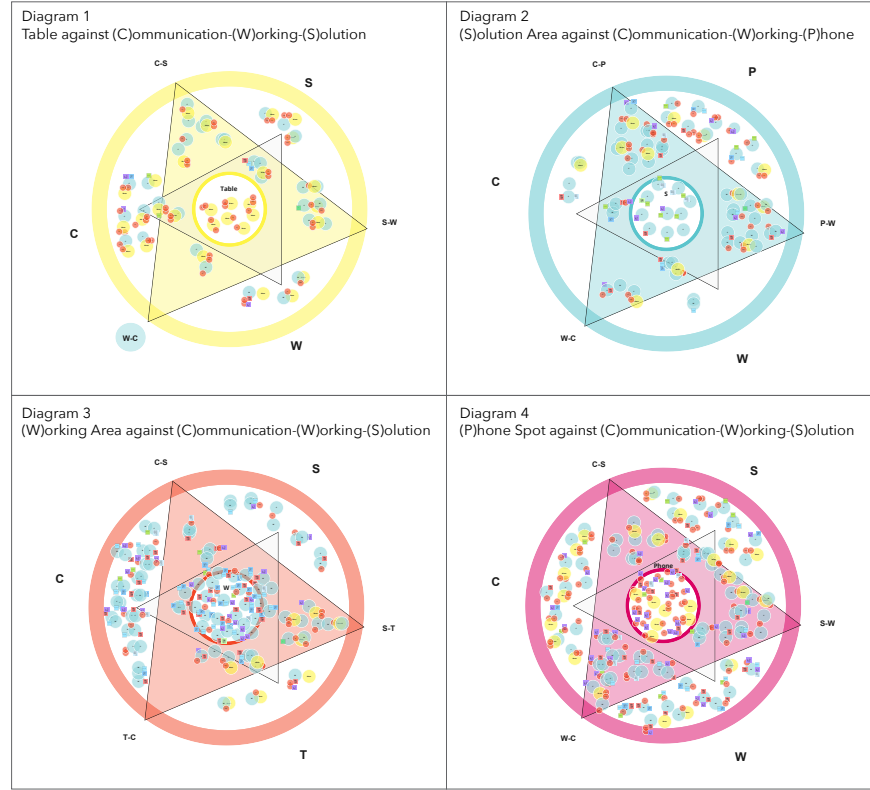


Fig. 4: Each diagram represents clustered spatial relationships between Tables, Phones, Communication Zones, Working Areas, and proposed Solution Points (S) for a new communication device.

“continuous visual connection.” Their concepts evoke *viewpoint augmentation*: cameras capable of auto-tracking, gaze- or voice-steered framing, and adaptive switching between close-up and contextual views. Instead of demanding attention, these systems would quietly keep the communicative thread intact through task transitions—moving from chopping to stirring to plating without a single touch. Such solutions transform visual continuity from an effortful act of camera management into a form of embodied awareness.

Communication with People Across ideation sessions, participants reimagined kitchen communication as a negotiation between *social rhythm* and *task rhythm*. They emphasized the frustration of having to choose between maintaining conversation and staying focused on the meal. In response, their solutions prioritized *social fluency*: lightweight cues that visualize who has the floor, soft notifications

for turn-taking, or adaptive modes that differentiate between addressing the instructor and speaking privately to peers. This fine-grained control of presence reflects a shift from automation toward *communicative orchestration*—systems that understand tempo, intent, and context rather than just transmitting sound.

Solution Approach: Mute & Unmute Default muting, though practical, broke the natural flow of talk. Participants proposed “smart muting”: voice-toggled microphones, context-based noise filtering, and gesture-triggered unmute actions to restore spontaneity. Some imagined context-aware mixing that isolates voices from background appliance noise or channels speech selectively (“talk to instructor,” “talk to group”). These ideas represent a form of *auditory augmentation for social coordination*, where systems listen with the user’s body and task, dynamically routing audio to sustain fluid, embodied dialogue.

Beyond Verbal Communication The most speculative yet emotionally charged ideas reached beyond words, suggesting that communication in the kitchen is inherently multisensory and affective. Participants envisioned XR and ubiquitous systems that could translate mood, taste, and atmosphere into new communicative textures—where cooking becomes both conversation and performance. Read through an augmentation lens, these concepts propose expanding communicative bandwidth through sensory richness.

Solution Approach 1: Communication with Taste Participants imagined *edible signals*—flavors, textures, or plating gestures—that could mark celebration, gratitude, or learning milestones. These “messages through food” express *material communication*, linking sensory pleasure to social meaning. As design speculation, such ideas frame augmentation not as substitution but as amplification of cultural ritual and embodied expression.

Solution Approach 2: Augmented Cooking Experience Participants extended the idea of sensory augmentation into atmosphere. They described responsive environments where ambient light, projection, or music adjusts to the recipe, region, or collective pace of the group. This *situated ambience augmentation* synchronizes digital cues with bodily tempo—deepening the sense of shared presence while preserving concentration.

Solution Approach 3: Immersive and Ambient Cooking Experience Sound was frequently reimaged as a communicative substrate. Participants proposed enhancing or spatializing natural kitchen sounds—the hiss of oil, the rhythm of chopping—as *auditory beacons* for coordination, safety, and empathy. Hearing another’s cooking progress became a subtle yet powerful way to stay connected. Such approaches exemplify *awareness through augmentation*: systems that expand perception without demanding attention.

Table 2: Kano Analysis for each artifact. Values represent the Euclidean distance to the ideal centers defined by the Kano model. Smaller numbers indicates aligns stronger to the category. Bold cells indicates the smallest number of the feature and close scores are both emphasized which indicates hybrid profile; Attractive and Performance.

	A	O	M	I	Result
Kitchen Portal	0.199	0.304	0.563	0.514	Attractive
Kitchen Portal on Cooking	0.224	0.277	0.559	0.535	Attractive
Kitchen Annotations	0.070	0.435	0.644	0.480	Attractive
Immersive Kitchen	0.163	0.535	0.612	0.338	Attractive

4.3 User Requirements from VR Prototype Evaluation

In the final phase of the study, nine professional cooks experienced three virtual prototypes in VR and evaluated them through a survey.

Table 2 shows the Kano analysis results for each artifact. While all artifacts are classified as Attractive, Kitchen Annotations exhibits the shortest distance, indicating the strongest alignment with this category. The Attractive category indicates that features are exciting for users but their absence does not cause dissatisfaction. This is understandable since these artifacts do not exist in today’s kitchen and participants only used them briefly in a virtual environment, so they did not develop a strong connection with them.

Conversely, Kitchen Portal on Cooking is technically classified as Attractive, but its high proximity to the Performance coordinate indicates a hybrid profile. Among all evaluated artifacts, it sits closest to the Performance threshold, indicating that users are excited by it but its absence would cause dissatisfaction especially during cooking. Some features were not experienced directly during the session; instead, they were presented as hypothetical features. Table 3 reports each feature’s Euclidean distance to the ideal points and highlights hybrid classifications in red.

The results show that privacy features on the Kitchen Portal, such as turning the camera on/off and the recording status indicator, as well as the privacy feature on Kitchen Annotations, cluster closer to Performance, indicating that their absence is likely to cause dissatisfaction. For the Kitchen Portal, Portability is also classified as a Performance feature, and Gesture Control emerges as a hybrid case. In contrast, Parallax Interaction appears as a hybrid between Indifferent and Attractive, suggesting it is comparatively low priority: participants do not see it as essential, and its absence would not be particularly upsetting. For Kitchen Annotations, features are predominantly Attractive, especially those supporting richer communication and message management such as 3D avatar messages, video messages, and archiving.

In summary, the Kano analysis suggests that privacy controls are near-essential requirements, and participants are most excited by communicating through 3D

avatar and video in both synchronous and asynchronous modes, which they perceive as particularly valuable during cooking.

Table 3: Kano analysis of Portal’s features. Values indicate Euclidean distance to the ideal Kano category centers: Attractive (A), One-Dimensional/Performance (O), Must-Be (M), and Indifferent (I). Features with similarly close distances to two categories are treated as hybrids.

Categories	Feature	A	O	M	I
Interactions and Indicators	Spatial Cues: Pointing to objects	0.170	0.382	0.538	0.416
	Parallax Interaction	0.308	0.410	0.407	0.304
	Privacy: Turning off the camera/video	0.398	0.113	0.468	0.604
	Gesture Control: Hand/eye tracking	0.292	0.302	0.425	0.419
	Voice Command: Voice control	0.224	0.386	0.483	0.367
	Recording Status: Status indicator	0.259	0.288	0.471	0.454
Mobility and Placement	Portable	0.305	0.195	0.536	0.585
	Placement behind the countertop	0.059	0.492	0.659	0.442
	Wall-mounted placement	0.186	0.393	0.521	0.389
	Placement on top of the cabinet	0.276	0.573	0.550	0.224
	Placement on the refrigerator	0.322	0.683	0.644	0.228
Form	Circular Portal	0.428	0.796	0.710	0.234
	Square Portal	0.399	0.596	0.458	0.118

5 Discussion

This study investigated how communication in active domestic workspaces, particularly kitchens, can be augmented through user-centered design and emerging XR technologies. Through three sequential stages, we identified communication challenges in remote kitchen interactions, explored user-driven design visions, and evaluated XR prototypes to assess user requirements. Our findings, structured across four observation-based challenges and four ideation-based thematic categories, extend existing literature and open speculative design pathways for spatial computing and XR applications in everyday life.

From Challenges to Design Insights The observational study revealed four key challenges: *Visibility*, *Vision*, *Audibility*, and *Device Interaction*, reflecting the embodied, multisensory, and often chaotic nature of the kitchen as an active workspace. Participants struggled with camera placement, screen limitations, appliance noise, and interaction breakdowns due to messy or occupied hands, challenges not merely technical but deeply entangled with spatial and bodily constraints, highlighting the inadequacy of conventional screen-based, touch-reliant

Table 4: Kano analysis of Kitchen Annotations’ and Immersive Kitchen features. Values indicate Euclidean distance to the ideal Kano category centers: Attractive (A), One-Dimensional/Performance (O), Must-Be (M), and Indifferent (I).

Feature	A	O	M	I
Kitchen Annotations				
3D Avatar / Object	0.097	0.424	0.612	0.453
Video message from the kitchen	0.094	0.453	0.615	0.427
3D avatar from the kitchen	0.101	0.420	0.608	0.451
Message from a mobile phone	0.222	0.323	0.498	0.439
Saving / Storing messages	0.113	0.393	0.609	0.479
Hiding messages / Privacy	0.338	0.203	0.437	0.514
Moving / Carrying messages	0.243	0.330	0.470	0.413
Immersive Kitchen				
Ambient Light	0.150	0.403	0.557	0.413
Theme Music	0.140	0.391	0.571	0.439
Ambient Images	0.155	0.448	0.561	0.371
Ambient Chef	0.158	0.342	0.605	0.523
Haptic Feedback	0.175	0.347	0.547	0.457
Audio Alerts / Cues	0.188	0.322	0.554	0.488

interfaces. The ideation workshops responded with speculative proposals, including wearable or spatially embedded communication tools (Theme 2: *Placement & Attributes*) and hybrid input modalities like voice commands and continuous camera tracking, aligning with Ju’s notion of Active Workspaces [11]. *Visibility* and *Vision* map to perceptual augmentation (e.g., multi-perspective capture informed by [10,18]), *Audibility* to auditory augmentation (noise-aware, role-sensitive capture as in [18]), and *Device Interaction* to hands-free, ubiquitous control consistent with kitchen-focused spatial systems [14,13]. The Kano analysis further confirmed these directions, with privacy controls emerging as near-essential and 3D avatar communication as the most exciting feature. Workshop ideas such as the “Always With Me” device extend this trajectory by suggesting lightweight XR or distributed display/audio arrangements that sustain communication without interrupting action, an augmentation-first stance rather than a mere usability fix.

Communication as a Situated Social Practice Beyond technical constraints, our findings underscore that communication in kitchens is deeply social. Theme 3 (*Communication with People*) captures how users desired systems supporting not only task coordination but also spontaneous conversation, shared experiences, and emotional resonance. The role of the workshop organizer as an ad-hoc facilitator who maintained group flow and mediated muting/unmuting revealed a critical gap in current HCI literature: communication management interfaces for synchronous, social use in active settings, where social fluency requires surfaces for lightweight turn-taking and selective addressing, especially when bodies and tools are in

motion. Prior systems like *Living Cookbook* [27] and *First Person Cooking* [16] illustrate the value of visual augmentation during cooking, yet do not explicitly handle the social orchestration seen in our workshops. Our findings close this gap by exposing fine-grained interactional dynamics, such as who needs help, who is plating, and who is waiting to speak, that can guide XR overlays and ubiquitous audio-visual cues toward augmented co-presence rather than simple instruction delivery.

Spatial Reasoning and Embodied Device Use Our kitchen layout analysis (Theme 1: *Spatial Insights*) revealed how users think through space when imagining new technologies. Communication tools were commonly associated with central, shared areas like tables, while intensive work areas (e.g., stove, sink) were treated as off-limits for tech placement, offering valuable insights for spatial computing: not all surfaces or zones in a kitchen are equal in their communicative affordances. The spatial diagrams created from participant-drawn plans serve as a methodological contribution, mapping embodied routines in a way that informs spatially adaptive XR interfaces where AR systems dynamically adjust notification positioning based on proximity to work or social zones. This orchestration extends the logic of projection- and sensing-based kitchens [14,18] and complements assistive spatial setups [13] by emphasizing augmentation of awareness and social continuity. While Knierim et al. [12] primarily addressed human–environment interaction, our findings expand this by showing how spatial reasoning and proxemic awareness underpin human–human communication itself, where layout, body orientation, and visibility shape the rhythm and success of collaboration, positioning the kitchen as a living testbed for *situated augmentation*: ubiquitous sensing and XR overlays that adapt to proxemic dynamics, respect task intensity, and sustain legibility among people, artifacts, and actions.

Speculative Futures of Multisensory Communication Theme 4 (*Beyond Verbal Communication*) pushes the boundary of what counts as communication in kitchens. From edible 3D-printed messages to ambient audio augmentation, participants envisioned ways of connecting through food, sound, and atmosphere, expanding the role of the kitchen into a site for affective, symbolic interaction that resonates with multisensory dining and remote togetherness explored in *Telelife* [19] and *CoDine* [32,31]. Dynamic lighting, projected ambience, or culturally themed soundscapes act as communicative scaffolds that deepen co-experience and synchronize group tempo, foregrounding augmentation of presence, feeling with and feeling there, rather than automation.

Towards Hybrid Interaction Together, these findings point to a framework for hybrid interaction in active domestic spaces, one that: (1) embraces spatial and bodily context as a core design input, (2) prioritizes low-friction multimodal control (e.g., voice, gesture, spatial audio), (3) enables ambient social awareness (e.g., subtle cues for turn-taking and attention), (4) blends asynchronous and synchronous modalities (e.g., quick voice snippets or presence markers), and (5) incorporates playful and sensory layers for richer co-experience. As work on smart,

sensing-rich kitchens [18,10] and domestic AR [12] has shown, everyday spaces are fertile ground for understanding embodied interaction. Our results suggest that augmenting communication, rather than solely instructing or automating, provides a transferable blueprint for other *Active Workspaces* [11], where ubiquitous computing and XR converge to enhance human perceptual, social, and communicative capabilities in daily life.

6 Conclusion

This study explored the challenges and opportunities of communication in kitchen environments, an exemplar of active workspaces, through a three-stage mixed-methods inquiry combining cooking workshop observations, co-design ideation sessions, and VR prototype evaluation. Addressing three research questions, we identified limitations in device placement, usability, and social interaction; translated user needs into three XR communication prototypes; and evaluated them using a Continuous Fuzzy Kano analysis. Our contributions include an empirical grounding for future kitchen-to-kitchen communication tools, a speculative design framework for XR-enabled domestic communication, and evidence that privacy controls are near-essential requirements in such systems. These findings frame communication not simply as information exchange but as a form of perceptual and embodied augmentation, enhancing awareness, coordination, and presence in motion.

This research has several limitations. The observational findings were not fully shared with ideation participants, preserving autonomy but limiting cross-pollination. Participants were not directed toward specific technologies during ideation, resulting in broader rather than XR-focused outcomes. Additionally, the study was conducted within a single cultural context, and findings may not generalize across diverse cultural or spatial configurations of kitchen use.

Future work should examine cross-cultural comparisons of communication behavior in active domestic workspaces and develop functional XR prototypes enabling hands-free, spatially aware, and multisensory interactions. Hybrid communication scenarios integrating asynchronous and synchronous modalities also warrant further investigation. The originality of this work lies in reframing the kitchen from a site of automation and assistance into a living laboratory for human-to-human communication and embodied augmentation. By extending the discourse of human augmentation into the social and communicative realm, this paper contributes a perspective that emphasizes augmentation not only of function, but of relation, expression, and shared experience.

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